

5 (**short**) A spherical thin balloon expands from volume $V_1 = 1$ litre to $V_2 = 5$ litres by the addition of $Q = 560$ J of heat. The balloon material resists expansion of the surface area A , resulting in work done by the gas of $W = p_{\text{atm}}\Delta V + \sigma\Delta A$, where ΔV is the change in volume, σ is the surface tension and ΔA is the change in surface area of the balloon. Assume that the balloon is filled with air (taken as a perfect gas) initially at temperature $T_1 = 200$ K and $p_1 = 1$ bar, and expands against a constant atmospheric pressure of $p_{\text{atm}} = 1$ bar.

(a) Calculate the mass of air within the balloon. [4]

(b) Calculate the final temperature when the surface tension of the balloon is $\sigma = 100 \text{ N m}^{-1}$. [6]